Failure-Aware Evaluation for Enterprise RAG Model Selection

Anonymous ACL submission

Abstract

Retrieval-Augmented Generation (RAG) systems are typically evaluated with aggregate scores—accuracy, relevance, faithfulness, citation accuracy—that obscure why a system fails under retrieval imperfections. We introduce a benchmark-driven evaluation methodology for characterizing retrieval-failure robustness, pairing clean and corrupted retrieval contexts for the same question across six controlled failure types. Using paired data from RAGFailBench across three benchmark seeds (206 paired questions, 9,270 judged instances), we evaluate five LLMs, each confirmed to run under identical configuration in every seed: DeepSeek V3.2, GPT-OSS, Llama 3.3, Nova Pro, and Qwen3-Next-80B-A3B. All five score within a narrow 0.98–1.00 band on clean contexts. Four then diverge sharply but stably once failures are introduced: Llama 3.3 resists hard negatives only 38.5% of the time (±2.9 points across seeds; pooled McNemar test, n=206, p<0.001), and Nova Pro has the highest clean accuracy but the weakest, most stable conflict-resolution score (71.8%, ±0.1 points). A fifth model shows substantial cross-seed instability, reported here as an open observation. Aggregate RAG metrics conceal robustness differences that are not sampling noise, and models can show cross-seed instability invisible to single-seed evaluation.

1 Introduction

Existing RAG evaluations report aggregate metrics — answer accuracy, relevance, faithfulness, citation accuracy — as a single score per model, implicitly assuming that all retrieval failures contribute equally to that score. We show this assumption is false: models that appear nearly identical under clean retrieval exhibit substantially different robustness under controlled retrieval failures, and a model's aggregate rank is a poor predictor of its rank on any individual failure type.

> ***Central claim: models that appear nearly identical under clean retrieval exhibit substantially different robustness under controlled retrieval failures, and this must be measured directly — aggregate accuracy will not reveal it.***

Retrieval-Augmented Generation grounds LLM outputs in external knowledge and is increasingly used for enterprise question answering, but retrieval rarely returns a single, sufficient passage: contexts may omit supporting evidence, contain contradictory passages, include a highly relevant but factually wrong distractor, fragment evidence across chunk boundaries, add irrelevant noise, or place evidence at an unfavorable position in a long context. These conditions stress different model capabilities — abstention, evidence arbitration, distractor resistance, cross-chunk reasoning, noise filtering, position robustness — and a model can be strong on one axis while weak on another even when its aggregate accuracy looks unremarkable next to competitors.

Several of these conditions are, at their core, instances of retrieval-induced uncertainty rather than gaps in a model's parametric knowledge: missing evidence requires recognizing that the model's epistemic state does not support an answer and abstaining accordingly — a calibration and selective-prediction problem — while conflict requires arbitrating between two sources under genuine ambiguity about which is correct, and evidence position tests whether a model's confidence in evidence is invariant to where that evidence appears in context. Characterizing how models behave under these conditions is therefore a study of uncertainty introduced by imperfect retrieval, distinct from and complementary to uncertainty arising from the question itself or from the model's parametric knowledge, and it connects directly to work on hallucination and knowledge-conflict resolution under uncertainty (Section 2). The cross-seed instability we report in Section 5.3 is itself a further form of uncertainty — variability in a model's own behavior across repeated, independently drawn evaluations — that is orthogonal to, and not resolved by, better calibration on any single evaluation run.

We present a failure-aware evaluation framework for evidence-based RAG model selection: for every benchmark question we construct a clean context and six failure-conditioned variants that isolate one retrieval failure at a time, then compare model behavior across conditions on the same question. This paired design attributes accuracy changes to a specific failure type rather than to question difficulty.

Our contributions are:

- A failure-aware evaluation framework that pairs clean and failure-conditioned retrieval contexts for the same question across six controlled failure types, supporting evidence-based RAG model selection rather than a single leaderboard score.
- A three-seed evaluation (206 paired questions, 9,270 judged instances) of five representative LLMs, each confirmed to run under identical configuration in every seed, combining standard RAG metrics with six failure-specific robustness metrics and a joint latency/cost analysis.
- A cross-seed stability analysis showing that four models are highly consistent across seeds (std ≤ 0.035 on nearly every failure type); a fifth model instead shows substantial cross-seed instability, which we report as an open observation (Appendix E) rather than a central finding.

We organize the study around four research questions: (RQ1) How do representative LLMs perform under different controlled retrieval failures? (RQ2) Do conventional aggregate metrics adequately capture robustness under retrieval failures? (RQ3) What distinct failure profiles emerge across different LLMs? (RQ4) How can failure-aware evaluation support practical model selection for enterprise RAG systems?

2 Related Work

Automatic RAG evaluation frameworks — RAGAS (Es et al., 2024), ARES (Saad-Falcon et al., 2024), RGB (Chen et al., 2024), and RAGTruth (Niu et al., 2024) — score answers under one retrieval condition and are not designed to compare a model across systematically varied failure types on paired questions. Large benchmarks — KILT (Petroni et al., 2021), Natural Questions (Kwiatkowski et al., 2019), TriviaQA (Joshi et al., 2017), HotpotQA (Yang et al., 2018), MuSiQue (Trivedi et al., 2022), and MultiHop-RAG (Tang and Yang, 2024) — evaluate a single retrieval condition per question and so cannot isolate a failure mode from question difficulty. Work on hallucination (Huang et al., 2025), adversarial retrieval corruption (Yoran et al., 2024; Zou et al., 2025), knowledge conflicts (Longpre et al., 2021; Xie et al., 2024; Xu et al., 2024), and long-context position effects (Liu et al., 2024) — much of it concerned, like this paper, with epistemic uncertainty and calibrated abstention — typically manipulates the prompt, corpus, or reasoning task in isolation rather than applying a unified taxonomy. Our framework builds on dense and retrieval-augmented generation (Karpukhin et al., 2020; Khattab and Zaharia, 2020; Guu et al., 2020; Izacard and Grave, 2021; Lewis et al., 2020; Shi et al., 2023; Trivedi et al., 2023; Asai et al., 2024; Sarthi et al., 2024; Thakur et al., 2021) and the pretrained-model foundations (Vaswani et al., 2017; Devlin et al., 2019; Brown et al., 2020; OpenAI, 2023) that RAG pipelines wrap, surveyed by Gao et al. (2023), but sits downstream, at the evaluation layer.

RGB is the closest prior system to ours in intent — it perturbs retrieval with noise and counterfactual evidence — but reports two failure conditions rather than a full six-type taxonomy and does not pair every instance with a clean control on the same question. Table 1 summarizes this positioning along two dimensions: whether a framework reports an aggregate quality score, and whether it reports failure-specific behavior on controlled, paired retrieval corruptions.

Work	Aggregate metrics	Failure-specific evaluation	Paired clean/failure design
RAGAS (Es et al., 2024)	✓	✗	✗
ARES (Saad-Falcon et al., 2024)	✓	✗	✗
RGB (Chen et al., 2024)	Partial	Noise + counterfactual only	✗
KILT / HotpotQA	✓	✗	✗

Work	Aggregate metrics	Failure-specific evaluation	Paired clean/failure design
Ours (RAGFailBench-based)	✓	Six controlled failure types	✓

Table 1: Positioning of this work relative to prior RAG evaluation frameworks and benchmarks.

3 Failure-Aware Evaluation Framework

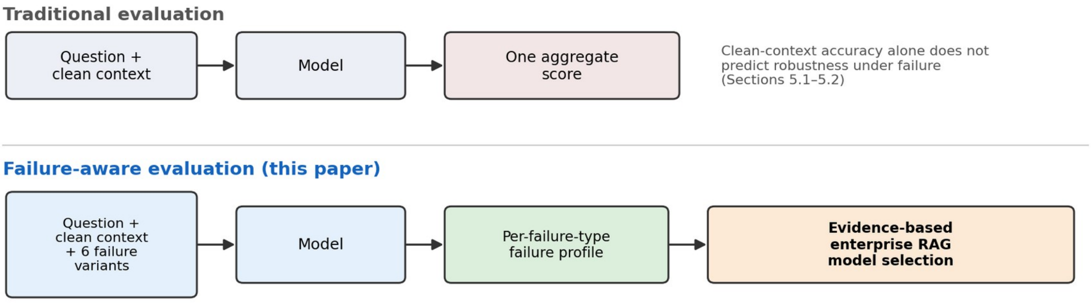

Figure 1: Traditional RAG evaluation collapses a system to one aggregate score; failure-aware evaluation instead runs every model on the same question under six controlled failure conditions and reports a per-failure-type profile, which is what supports evidence-based enterprise model selection (Section 6).

The framework has three stages. First, RAGFailBench constructs, for each seed question, a clean retrieval context together with six controlled failure variants. Second, every model is run under an identical prompt and decoding configuration across the clean condition and all six failure conditions for the same question. Third, we compute standard quality metrics on the clean condition and failure-specific robustness metrics that compare each model's clean and failure-conditioned behavior, producing a per-model failure profile rather than a single leaderboard score (the detailed eight-stage pipeline diagram is in Appendix F).

Failure type	Description	Expected behavior
Missing evidence	Supporting evidence removed from the context	Abstain
Conflict	A contradictory passage is inserted alongside the gold evidence	Answer with gold evidence
Hard negative	A highly relevant but factually incorrect distractor replaces the gold evidence	Abstain
Boundary	The supporting evidence is fragmented across chunk boundaries	Answer, citing the split chunks
Noise	Irrelevant passages are injected alongside the gold evidence	Answer, unaffected by noise
Evidence position	The gold evidence is relocated to the first, middle, or last position in the context	Answer regardless of position

Table 2: Six controlled retrieval failure types and the expected model behavior under each.

Concretely, for each seed question the pipeline draws a clean context and six paired failure contexts from RAGFailBench, issues an identical prompt to every model under fixed decoding settings, scores each response with rule-based checks and two independent LLM judges, and aggregates the paired clean-versus-failure results into the metrics below. On the clean condition we report four standard metrics: Answer Accuracy, Relevance, Faithfulness, and Citation Accuracy. On the six failure conditions we compute established failure-specific robustness scores rather than proposing new metrics: Missing-Evidence Abstention Rate (MAR, correct abstentions over missing-evidence cases), Conflict Resolution Score (CRS, answers using gold evidence over conflict cases), Hard-Negative Resistance (HNR, correct abstentions over hard-negative cases), Noise Robustness Score (NRS, noise accuracy over clean accuracy), Boundary Robustness Score (BRS, boundary accuracy over clean accuracy), and Position Drop (first-position accuracy minus last-position accuracy, with middle-position accuracy as a lost-in-the-middle diagnostic). Full formulas are given in Appendix A.

4 Experimental Setup

We evaluate five models spanning open-weight, MoE, and proprietary managed deployments, accessed through a unified Bedrock-hosted inference interface with shared decoding settings (temperature = 0, top_p = 1.0, max_tokens = 256): Meta Llama 3.3 (Dubey et al., 2024), OpenAI GPT-OSS, DeepSeek V3.2 (DeepSeek-AI, 2024), Amazon Nova Pro (Amazon AGI, 2024), and Qwen3-Next-80B-A3B (Yang et al., 2025). We verified via raw per-request inference logs — not just the aggregated evaluation output — that all five models ran under an identical model identifier in every one of the three benchmark seeds; this check caught a stale model-identity label in our own evaluation output, corrected here and detailed in Appendix B. A sixth model, Claude Sonnet 5, is configured but not yet enabled pending Bedrock/Anthropic marketplace access approval.

Model	Category	Seed coverage
DeepSeek V3.2	Open-weight / MoE	s42, s123, s2026
OpenAI GPT-OSS	Open-weight	s42, s123, s2026
Meta Llama 3.3	Open-weight (large)	s42, s123, s2026
Amazon Nova Pro	Proprietary, cloud-native	s42, s123, s2026
Qwen3-Next-80B-A3B	Open-weight / MoE (large)	s42, s123, s2026

Table 3: Models evaluated, with confirmed identical configuration across all three benchmark seeds.

4.1 Benchmark: Source, Construction, and Seeds

Each question in RAGFailBench is extracted from a Wikipedia article (page ID, revision ID, and section identified at construction time), paired with a short factual question, a gold answer, and a gold supporting passage. The construction pipeline then builds, for every question, a clean context (the gold passage plus topically related distractor passages drawn from the same source pool) and six controlled failure variants that each perturb the clean context in exactly one way — removing the gold passage (missing evidence), inserting a contradictory passage (conflict), substituting a highly relevant but wrong passage (hard negative), splitting the gold passage across chunk boundaries (boundary), adding irrelevant passages (noise), or relocating the gold passage within the context (evidence position) — while holding the question, gold answer, and non-perturbed passages fixed.

We evaluate on three independently generated benchmark instances produced by this pipeline, using random seeds 42, 123, and 2026 while keeping the construction pipeline, source pool, and all other parameters fixed; the three seeds therefore differ only in which questions and distractor passages were sampled. We use three seeds, rather than one, specifically to distinguish structural model weaknesses that replicate across independently sampled question sets from single-draw artifacts that do not — the distinction that drives Section 5.3. Table 4 summarizes the three resulting benchmark instances; question categories (person, location, organization/product, historical event, science/technology) are approximately balanced within each seed.

Seed	Questions	Paired instances (all conditions)
Seed 42	75	600
Seed 123	67	536
Seed 2026	64	512

Table 4: The three benchmark instances used in this study. Each question contributes 8 paired rows (1 clean + 5 non-positional failures + 3 evidence-position placements); a paired instance count of 600 for seed 42 is 75 × 8.

Data are filtered to medium severity for five of the six failure types; evidence position retains all three placements (first/middle/last) as a severity proxy. All three benchmark seeds are fully evaluated, totaling 206 paired questions and 9,270 judged instances across the five models. Responses are checked with rule-based validators, then scored by two independent LLM judges (GPT-OSS and DeepSeek) following the LLM-as-judge paradigm (Zheng et al., 2023); judge agreement ranges from 75.6% to 81.9% across seeds, with the remainder queued for human review (Appendix E). Section 5.5 separately reports latency and cost from a streaming, concurrency-1 measurement pass over a 30-question subset of seed 42 (Appendix D gives the full methodology).

5 Results

5.1 RQ1/RQ2: Model Performance and Aggregate-Metric Validity

All five models score within a narrow 0.960–1.000 band on clean-context accuracy (seed 42; full table in Appendix C), providing almost no discriminative signal between them. Table 5 reports accuracy under each of the six controlled failure conditions for the same seed: the clustering seen on clean data disappears entirely, with a 69.3-point spread on conflict alone (0.280–0.973).

Model	Missing Ev.	Conflict	Hard Neg.	Boundary	Noise	Evid. Position
DeepSeek V3.2	0.840	0.973	0.893	0.920	0.933	0.960
GPT-OSS	0.760	0.947	0.867	0.787	0.973	0.973
Llama 3.3	0.747	0.973	0.347	0.907	1.000	0.960
Nova Pro	0.707	0.787	0.840	0.893	0.973	0.987
Qwen3-Next-80B-A3B	0.787	0.280	0.747	0.720	0.653	0.809

Table 5: Accuracy under each of the six controlled failure types, seed 42 (n = 75 questions, 225 for evidence position).

This single-seed pattern generalizes for four of the five models. Table 6 repeats the clean-versus-failure rank comparison using the pooled 3-seed mean of each model's failure accuracy (Section 5.3), for the four models whose scores are stable enough across seeds to support a meaningful rank; the fifth model is excluded here because its own cross-seed instability makes a single pooled rank misleading (full single-seed comparison in Appendix C).

Model	Clean acc. (3-seed)	Clean rank	Mean failure acc. (3-seed)	Failure rank	Rank change
DeepSeek V3.2	0.996	1	0.930	1	0
Nova Pro	0.990	2	0.871	3	−1
Llama 3.3	0.986	3	0.833	4	−1
GPT-OSS	0.981	4	0.901	2	+2

Table 6: Clean-accuracy rank versus pooled 3-seed mean failure-accuracy rank, for the four rank-stable models. Rank 1 = best.

GPT-OSS rises two places (4th to 2nd) once failures are introduced, while Llama 3.3 and Nova Pro each drop a place; DeepSeek V3.2 is the only model whose rank is unchanged. Section 5.4 confirms with McNemar tests (n = 206 pairs) and a bootstrap over the pooled questions that these gaps are not sampling noise (Appendix E gives full test statistics). This directly answers RQ2: conventional aggregate accuracy does not adequately capture robustness under retrieval failure, and a model's aggregate rank is not a reliable predictor of its rank on any individual failure type — a conclusion that holds on a single seed and strengthens when checked across three.

5.2 RQ3: Failure Profiles

Reading Table 5 with the failure-specific behavioral metrics (Appendix C, Table 10) produces five distinct failure profiles rather than one ranking. DeepSeek V3.2 is the most uniformly robust, staying at or above 0.84 accuracy on every failure type. Nova Pro has the highest clean accuracy (1.000) but is weakest on missing evidence (MAR = 0.707) and conflict (CRS = 0.720) — the highest aggregate score does not correspond to the most trustworthy behavior under failure. GPT-OSS is strong overall with a dip on boundary fragmentation (BRS = 0.808) despite perfect noise robustness (NRS = 1.000). Llama 3.3 is near-ceiling on five of six failure types but is fooled by a hard-negative distractor roughly two-thirds of the time (HNR = 0.347). The fifth model is least robust on several axes, though Section 5.3 shows this pattern does not replicate in the other two seeds (Appendix C.4 gives a representative instance).

5.3 Cross-Seed Stability

Sections 5.1–5.2 report a single seed. We verified via raw inference logs that all five models ran under identical configuration in every seed, so cross-seed differences reflect model behavior on different questions, not a change in which model was evaluated.

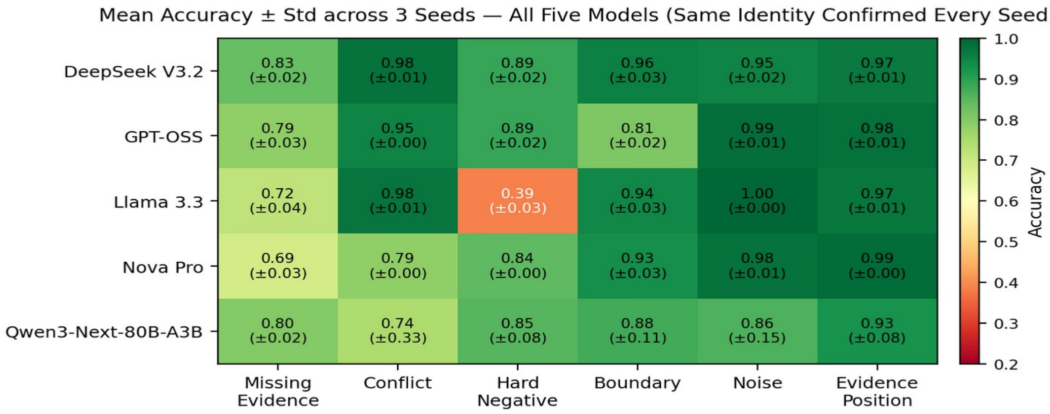

Figure 2: Mean accuracy ± standard deviation across three seeds, for all five models (confirmed identical configuration in every seed).

Four of five models are strikingly stable: every model×condition cell for these four has a cross-seed standard deviation at or below 0.035. Llama 3.3's hard-negative weakness is the clearest example (0.347, 0.418, 0.391 across seeds; mean 0.385, std 0.029; McNemar $\chi 2$ = 94.7–98.2, $p < 0.001$ against both DeepSeek and GPT-OSS, pooling all 206 paired questions). Nova Pro's conflict weakness is equally stable (CRS = 0.720, 0.716, 0.719; std = 0.001). Both graduate from single-seed observations to confirmed, replicated properties of these models within our benchmark. The fifth model instead shows substantial cross-seed instability on five of six failure types despite the same confirmed configuration in every seed (two-proportion test, $p < 0.001$ on five of six conditions). We ruled out API and parsing errors as the cause but do not have a confirmed mechanism; we report this as an open observation rather than a resolved finding, and give the full statistical treatment and a response-level analysis in Appendix E, since the main contribution of this paper is the evaluation framework itself, not this one model's behavior.

5.4 Statistical Significance

Because every model answers the same questions per condition, we can test directly whether the gaps above are sampling noise. McNemar's test (McNemar, 1947) on paired per-question correctness, pooled across all three seeds, confirms both headline gaps at $p < 0.001$ (Llama 3.3 vs. GPT-OSS and vs. DeepSeek V3.2 on hard negatives, $\chi 2$ = 94.7–98.2; Nova Pro vs. DeepSeek V3.2 on conflict and missing evidence, $\chi 2$ = 28.0–38.0). A bootstrap (Efron, 1979; 5,000 resamples over the 206 pooled questions) gives non-overlapping 95% confidence intervals for the four rank-stable models' mean failure accuracy (DeepSeek V3.2 minus Llama 3.3: [0.083, 0.120]; minus Nova Pro: [0.045, 0.075], both excluding zero), confirming the Table 6 ranking is not an artifact of any single seed's question draw. Full test tables are in Appendix E.

5.5 Quality-Latency-Cost Trade-off

Robustness is only one axis practitioners weigh. We measured end-to-end (E2E) latency, time-to-first-token (TTFT), and Bedrock per-token cost over a 30-question, concurrency-1 subset of seed 42 (4,050 requests, 99.93%

success; methodology and full per-model latency table in Appendix D). Table 7 places these figures alongside each model's seed-42 mean failure accuracy.

Model	Mean failure acc. (s42)	E2E p95 latency	Cost / 1K requests
DeepSeek V3.2	0.920	2,105 ms	$0.742
GPT-OSS	0.885	2,824 ms	$0.261
Nova Pro	0.865	1,076 ms	$0.994
Llama 3.3	0.822	1,340 ms	$0.834
Qwen3-Next-80B-A3B	0.666	1,386 ms	$0.216

Table 7: Robustness, tail latency, and cost together. Qwen3-Next-80B-A3B's accuracy is the seed-42 value only; Section 5.3 shows it is not representative of the other two seeds (0.936, 0.932).

No model wins on all three axes: DeepSeek V3.2 has the strongest, most stable robustness but is neither fastest nor cheapest; Nova Pro has the most stable tail latency but the highest cost; GPT-OSS is inexpensive but slowest; Qwen3-Next-80B-A3B is cheapest but its accuracy figure is exactly the one Section 5.3 shows does not replicate. Cost also varies sharply by condition, driven by each failure operator's context-construction design rather than a simple “noise adds tokens” relationship (Appendix D, Table 13).

6 Discussion: Implications for Enterprise Model Selection

Four findings follow from Sections 5.1–5.5. First, aggregate rank does not survive the introduction of failures: using the pooled 3-seed evidence, GPT-OSS rises from 4th to 2nd place and Llama 3.3 and Nova Pro each drop a place once the ranking criterion changes from clean accuracy to mean failure accuracy, confirmed by non-overlapping bootstrap intervals and pooled McNemar tests. Second, open-weight models are not uniformly weaker than the proprietary model here: DeepSeek V3.2 and GPT-OSS outperform Nova Pro on pooled mean failure accuracy despite Nova having the highest clean accuracy in every seed. Third, cross-seed replication separates structural weaknesses from single-run artifacts: Llama 3.3’s hard-negative weakness and Nova Pro’s conflict weakness both replicate almost exactly across seeds, while one model’s seed-42 weaknesses did not replicate despite confirmed identical configuration — multi-seed evaluation is a reliability check on the model, not only the benchmark. Fourth, no model is universally best: each rank-stable model has at least one failure type on which it is weakest, so “best RAG model” is only well-defined once the target failure distribution is specified.

The practical implication is that the right model depends on which retrieval failures and deployment constraints are most likely. An enterprise QA system exposed mainly to missing-evidence and conflict failures should weigh DeepSeek V3.2’s stable advantage over Nova Pro’s stable weakness on both, despite Nova’s higher clean accuracy. A customer-support system over noisy, loosely chunked documentation is more exposed to noise and boundary failures, where GPT-OSS’s stable noise robustness is relevant — though its slow tail latency (Table 7) may rule it out for latency-sensitive deployments. A system that cannot tolerate a hard-negative distractor should treat Llama 3.3’s replicated weakness there as a confirmed deployment risk, not a single-run fluke. We do not recommend a single best model: practitioners should weight the failure-specific scores in this paper by their own failure modes, latency, and cost constraints, using clean-context accuracy only as a baseline sanity check.

7 Conclusion

Models that appear nearly identical under clean retrieval exhibit substantially different robustness under controlled retrieval failures. Across three independent benchmark seeds, four of five models with confirmed identical configuration are nearly indistinguishable on clean accuracy (0.981–0.996) yet diverge by up to 21 points on pooled mean failure accuracy, with the two headline gaps confirmed by McNemar tests ($p < 0.001$) and non-overlapping bootstrap intervals. A fifth model showed substantial cross-seed instability we could not attribute to API or configuration differences, reported here as an open observation. Aggregate RAG metrics alone are insufficient for model selection and should be reported alongside failure-conditioned, cross-seed evidence.

8 Limitations

The following state the boundaries of what this study supports and what would need to change to extend each claim; each maps to a concrete direction for future work rather than a generic caveat.

- Representative, not exhaustive, model coverage. Five configurations (a sixth, Claude Sonnet 5, is configured but not enabled) spanning open-weight, MoE, and proprietary deployments; findings should not be assumed to generalize to untested families (e.g., Gemini, Claude, Mistral).
- Failure taxonomy scope. Six controlled failure types common in enterprise RAG; temporal knowledge drift, multi-hop retrieval failures, multilingual retrieval, and adversarial prompt interactions are out of scope.
- Controlled rather than fully production retrieval failures. Each condition is injected systematically and in isolation at medium severity; real systems may exhibit multiple simultaneous failure modes this benchmark does not yet compose, and all data derive from one benchmark construction pipeline.
- Predominantly LLM-judge evaluation with sampled human verification. Judges agree on 75.6–81.9% of instances; the remainder (≈1,902 across seeds) are flagged for human review, of which 926 have been drawn into a review sample and adjudication is not yet complete for all of them.
- Unresolved model-specific variation. Qwen3-Next-80B-A3B's cross-seed instability (Section 5.3) is not attributable to API errors, parsing failures, or configuration differences that we could test; whether it reflects genuine model behavior or serving-infrastructure characteristics is an open question.
- Application-specific deployment factors not jointly optimized. The Section 5.5 measurements cover 30 of 75 seed-42 questions at concurrency = 1 and a single pricing date (2026-08-06); they were not repeated on seeds 123/2026 or under concurrent load, and a practitioner-facing selection tool would need to combine robustness, latency, cost, and compliance jointly rather than defaulting to whichever model scores highest on one axis.

References

Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2024. Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection. In The Twelfth International Conference on Learning Representations (ICLR). https://openreview.net/forum?id=hSyW5go0v8

Amazon AGI. 2024. The Amazon Nova Family of Models: Technical Report and Model Card. Amazon Technical Reports. https://arxiv.org/abs/2506.12103

Anonymous. RAGFailBench: Automated Construction and Controllable Failure-Injection Benchmarking for Retrieval-Augmented Generation. Repository link omitted for anonymous review; will be released upon publication.

Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, et al. 2020. Language Models are Few-Shot Learners. In Advances in Neural Information Processing Systems 33 (NeurIPS). https://proceedings.neurips.cc/paper/2020/hash/1457c0d6bfcb4967418bfb8ac142f64a-Abstract.html

Jiawei Chen, Hongyu Lin, Xianpei Han, and Le Sun. 2024. Benchmarking Large Language Models in Retrieval-Augmented Generation. In Proceedings of the AAAI Conference on Artificial Intelligence, volume 38, pages 17754–17762. https://doi.org/10.1609/aaai.v38i16.29728

DeepSeek-AI. 2024. DeepSeek-V3 Technical Report. arXiv:2412.19437. https://arxiv.org/abs/2412.19437

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT). https://aclanthology.org/N19-1423/

Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, et al. 2024. The Llama 3 Herd of Models. arXiv:2407.21783. https://arxiv.org/abs/2407.21783

Bradley Efron. 1979. Bootstrap Methods: Another Look at the Jackknife. The Annals of Statistics, 7(1):1–26. https://doi.org/10.1214/aos/1176344552

Shahul Es, Jithin James, Luis Espinosa Anke, and Steven Schockaert. 2024. RAGAs: Automated Evaluation of Retrieval Augmented Generation. In Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations, pages 150–158. https://aclanthology.org/2024.eacl-demo.16/

Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, and Haofen Wang. 2023. Retrieval-Augmented Generation for Large Language Models: A Survey. arXiv:2312.10997. https://arxiv.org/abs/2312.10997

Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, and Ming-Wei Chang. 2020. REALM: Retrieval-Augmented Language Model Pre-Training. In Proceedings of the 37th International Conference on Machine Learning (ICML). https://proceedings.mlr.press/v119/guu20a.html

Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, and Ting Liu. 2025. A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions. ACM Transactions on Information Systems, 43(2). https://doi.org/10.1145/3703155

Gautier Izacard and Edouard Grave. 2021. Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering. In Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics (EACL), pages 874–880. https://aclanthology.org/2021.eacl-main.74/

Mandar Joshi, Eunsol Choi, Daniel S. Weld, and Luke Zettlemoyer. 2017. TriviaQA: A Large Scale Distantly Supervised Challenge Dataset for Reading Comprehension. In Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (ACL), pages 1601–1611. https://aclanthology.org/P17-1147/

Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. Dense Passage Retrieval for Open-Domain Question Answering. In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), pages 6769–6781. https://aclanthology.org/2020.emnlp-main.550/

Omar Khattab and Matei Zaharia. 2020. ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT. In Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval. https://doi.org/10.1145/3397271.3401075

Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, et al. 2019. Natural Questions: A Benchmark for Question Answering Research. Transactions of the Association for Computational Linguistics, 7:452–466. https://aclanthology.org/Q19-1026/

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In Advances in Neural Information Processing Systems 33 (NeurIPS), pages 9459–9474. https://proceedings.neurips.cc/paper/2020/hash/6b493230205f780e1bc26945df7481e5-Abstract.html

Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. Lost in the Middle: How Language Models Use Long Contexts. Transactions of the Association for Computational Linguistics, 12:157–173. https://aclanthology.org/2024.tacl-1.9/

Shayne Longpre, Kartik Perisetla, Anthony Chen, Nikhil Ramesh, Chris DuBois, and Sameer Singh. 2021. Entity-Based Knowledge Conflicts in Question Answering. In Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP). https://aclanthology.org/2021.emnlp-main.565/

Quinn McNemar. 1947. Note on the Sampling Error of the Difference between Correlated Proportions or Percentages. Psychometrika, 12(2):153–157. https://doi.org/10.1007/BF02295996

Cheng Niu, Yuanhao Wu, Juntong Zhu, Siliang Xu, Kai Shum, Randy Zhong, Juntong Song, and Tong Zhang. 2024. RAGTruth: A Hallucination Corpus for Developing Trustworthy Retrieval-Augmented Language Models. In Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL), pages 10862–10878. https://aclanthology.org/2024.acl-long.585/

OpenAI. 2023. GPT-4 Technical Report. arXiv:2303.08774. https://arxiv.org/abs/2303.08774

Fabio Petroni, Aleksandra Piktus, Angela Fan, Patrick Lewis, et al. 2021. KILT: a Benchmark for Knowledge Intensive Language Tasks. In Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT). https://aclanthology.org/2021.naacl-main.200/

Jon Saad-Falcon, Omar Khattab, Christopher Potts, and Matei Zaharia. 2024. ARES: An Automated Evaluation Framework for Retrieval-Augmented

Generation Systems. In Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT), pages 338–354. https://aclanthology.org/2024.naacl-long.20/

Parth Sarthi, Salman Abdullah, Aditi Tuli, Shubh Khanna, Anna Goldie, and Christopher D. Manning. 2024. RAPTOR: Recursive Abstractive Processing for Tree-Organized Retrieval. arXiv:2401.18059. https://arxiv.org/abs/2401.18059

Weijia Shi, Sewon Min, Michihiro Yasunaga, Minjoon Seo, Rich James, Mike Lewis, Luke Zettlemoyer, and Wen-tau Yih. 2023. REPLUG: Retrieval-Augmented Black-Box Language Models. arXiv:2301.12652. https://arxiv.org/abs/2301.12652

Yixuan Tang and Yi Yang. 2024. MultiHop-RAG: Benchmarking Retrieval-Augmented Generation for Multi-Hop Queries. arXiv:2401.15391. https://arxiv.org/abs/2401.15391

Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. 2021. BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models. In Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks (NeurIPS). https://openreview.net/forum?id=wCu6T5xFjeJ

Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2022. MuSiQue: Multihop Questions via Single-hop Question Composition. Transactions of the Association for Computational Linguistics, 10:539–554. https://aclanthology.org/2022.tacl-1.31/

Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2023. Interleaving Retrieval with Chain-of-Thought Reasoning for Knowledge-Intensive Multi-Step Questions. In Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL), pages 10014–10037. https://aclanthology.org/2023.acl-long.557/

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. 2017. Attention Is All You Need. In Advances in Neural Information Processing Systems 30 (NeurIPS), pages 5998–6008. https://proceedings.neurips.cc/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html

Jian Xie, Kai Zhang, Jiangjie Chen, Renze Lou, and Yu Su. 2024. Adaptive Chameleon or Stubborn Sloth: Revealing the Behavior of Large Language Models in Knowledge Conflicts. In The Twelfth International Conference on Learning Representations (ICLR). https://openreview.net/forum?id=auKAUJZMO6

Rongwu Xu, Zehan Qi, Zhijiang Guo, Cunxiang Wang, Hongru Wang, Yue Zhang, and Wei Xu. 2024. Knowledge Conflicts for LLMs: A Survey. In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP), pages 8541–8565. https://aclanthology.org/2024.emnlp-main.486/

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, and 1 others. 2025. Qwen3 Technical Report. arXiv:2505.09388. https://arxiv.org/abs/2505.09388

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. 2018. HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering. In Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP), pages 2369–2380. https://aclanthology.org/D18-1259/

Ori Yoran, Tomer Wolfson, Ori Ram, and Jonathan Berant. 2024. Making Retrieval-Augmented Language Models Robust to Irrelevant Context. In The Twelfth International Conference on Learning Representations (ICLR). https://openreview.net/forum?id=ZS4m74kZpH

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In Advances in Neural Information Processing Systems 36 (NeurIPS). https://proceedings.neurips.cc/paper_files/paper/2023/hash/91f18a1287b398d378ef22505bf41832-Abstract-Datasets_and_Benchmarks.html

Wei Zou, Runpeng Geng, Binghui Wang, and Jinyuan Jia. 2025. PoisonedRAG: Knowledge Corruption Attacks to Retrieval-Augmented Generation of Large Language Models. In 34th USENIX Security Symposium, pages 3827–3844. https://www.usenix.org/conference/usenixsecurity25/presentation/zou-poisonedrag

39 of the 40 entries above include a verified DOI, ACL Anthology, OpenReview, arXiv, or publisher link (the RAGFailBench entry is anonymized for review and will be linked upon publication), each confirmed against the publisher or archive record during drafting; author lists longer than seven are truncated with "et al." Final formatting should still be checked against the target venue's citation style before submission.

Appendix

This appendix gives full derivations, formulas, and supplementary tables for results summarized in the main text. It is not required to follow the paper's central argument.

A. Failure-Specific Metric Formulas

- MAR = (# correct abstentions) / (# missing-evidence cases)
- CRS = (# answers using gold evidence) / (# conflict cases)
- HNR = (# correct abstentions) / (# hard-negative cases)
- NRS = Accuracy(noise) / Accuracy(clean)
- BRS = Accuracy(boundary) / Accuracy(clean)
- Position Drop = Accuracy(first) − Accuracy(last), with Accuracy(middle) reported alongside as a lost-in-the-middle diagnostic.

B. Data-Hygiene Note

While preparing the cross-seed analysis (Section 5.3) we found that the seed-42 evaluation output (results/eval/s42/evaluated.jsonl) labeled the fifth model "Qwen/Qwen2.5-7B-Instruct," while the seed-123 and seed-2026 outputs labeled it "qwen3-next-80b-a3b." We traced this to the raw, pre-evaluation inference logs (results/s42/*/qwen.jsonl), which record the model actually invoked at request time for every row: every seed-42 request, across all six failure conditions, was sent to qwen3-next-80b-a3b, matching configs/models.yaml and matching seeds 123 and 2026. The "Qwen2.5-7B-Instruct" string appears only in the downstream evaluation join and matches the label used in an earlier 20-question pilot run, the most likely source of the stale label. We treat all five models as having identical configuration in every seed, correct the label throughout this paper, and flag the mislabeled evaluation output as a pipeline bug to fix independently of this analysis.

C. Seed-42 Supplementary Tables and Figures

C.1 Clean-Condition Quality Metrics

Model	Accuracy	Relevance	Faithfulness	Citation Acc.
DeepSeek V3.2	0.987	1.000	1.000	1.000
GPT-OSS	0.973	0.967	0.973	1.000
Llama 3.3	0.973	0.987	0.987	0.987
Nova Pro	1.000	0.993	0.993	1.000
Qwen3-Next-80B-A3B	0.960	0.933	0.947	1.000

Table 8: Clean-condition quality metrics, seed 42 (n = 75 questions per model).

C.2 Single-Seed Rank Comparison

Model	Clean accuracy	Clean rank	Mean failure accuracy	Failure rank	Rank change
Nova Pro	1.000	1	0.865	3	−2
DeepSeek V3.2	0.987	2	0.920	1	+1
GPT-OSS	0.973	3 (tie)	0.885	2	+1
Llama 3.3	0.973	3 (tie)	0.822	4	−0.5
Qwen3-Next-80B-A3B	0.960	5	0.666	5	0

Table 9: Clean-accuracy rank versus mean failure-accuracy rank, seed 42 only. Rank 1 = best.

C.3 Failure-Specific Behavioral Metrics and Evidence Position

Model	MAR	CRS	HNR	NRS	BRS	NoiseDrop	PosDrop
DeepSeek V3.2	0.840	0.920	0.893	0.946	0.932	0.053	0.027

Model	MAR	CRS	HNR	NRS	BRS	NoiseDrop	PosDrop
GPT-OSS	0.760	0.893	0.867	1.000	0.808	0.000	0.000
Llama 3.3	0.747	0.920	0.347	1.027	0.932	−0.027	0.040
Nova Pro	0.707	0.720	0.840	0.973	0.893	0.027	−0.013
Qwen3-Next-80B-A3B	0.787	0.227	0.747	0.681	0.750	0.307	0.173

Table 10: Failure-specific behavioral metrics, seed 42.

Model	First	Middle	Last
DeepSeek V3.2	0.987	0.933	0.960
GPT-OSS	0.973	0.973	0.973
Llama 3.3	0.973	0.973	0.933
Nova Pro	0.973	1.000	0.987
Qwen3-Next-80B-A3B	0.987	0.627	0.813

Table 11: Accuracy by gold-evidence position, seed 42 (n = 75 per position per model).

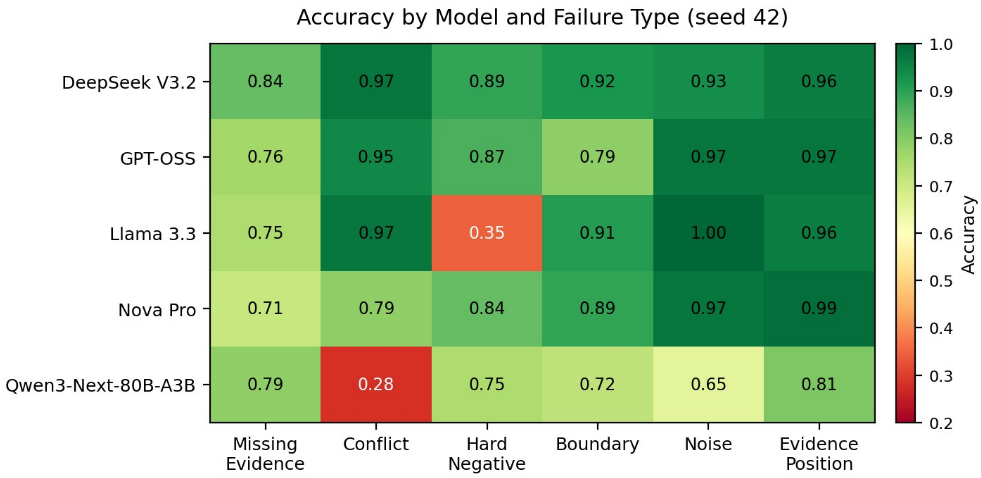

Figure 3: Heatmap of accuracy by model (rows) and failure type (columns), seed 42.

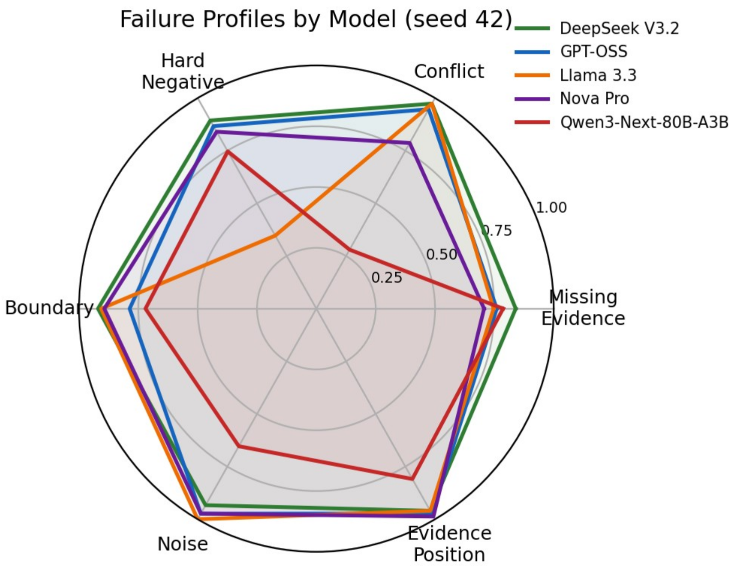

Figure 4: Radar chart of per-model accuracy across the six failure types, seed 42.

C.4 Case Study: One Conflict Instance

The seed-42 instance seed_000003 asks: "When did Rudolf Ludwig Mössbauer die?" (gold answer: 14 September 2011). The context contains a gold chunk stating "...31 January 1929 – 14 September 2011) was a German physicist who shared the 1961 Nobel Prize..." and an injected conflicting chunk stating "...31 January 1929 – 14 September 2008) was a German physicist who shared the 1961 Nobel Prize..." — near-identical except for the date. Qwen3-Next-80B-A3B answered "14 September 2008," exactly matching the injected chunk. This is representative rather than exceptional: of the 54 seed-42 conflict instances where Qwen3-Next-80B-A3B answered incorrectly, 48 (88.9%) reproduced the injected alternate answer exactly, one abstained, and 5 matched neither value. The errors are fluent, well-formed answers sourced from the wrong chunk, not garbled or refused output, consistent with the conflict operator constructing an injected chunk that is topically and stylistically indistinguishable from the gold chunk except for the disputed fact.

D. Latency and Cost Methodology and Detail

We ran a streaming, concurrency-1 measurement pass over 30 of the 75 seed-42 questions across all nine context conditions (clean; five non-positional failure types; first/middle/last evidence position), all five models, and 3 repetitions per case, preceded by 5 untimed warm-up calls per model: $30 \times 9 \times 5 \times 3 =$ 4,050 timed requests, recording E2E latency, TTFT, token counts, and Bedrock per-token cost (us-east-1, standard tier, priced as of 2026-08-06). 4,047 of 4,050 requests (99.93%) succeeded. We confirmed the fifth model's identity from a per-request exact_model_id field (= qwen3-next-80b-a3b for every "qwen" record).

Model	E2E p50	E2E p95	TTFT p50	Cost / 1K requests
Llama 3.3	589 ms	1,340 ms	377 ms	$0.834
Nova Pro	675 ms	1,076 ms	469 ms	$0.994
Qwen3-Next-80B-A3B	832 ms	1,386 ms	631 ms	$0.216
DeepSeek V3.2	939 ms	2,105 ms	661 ms	$0.742
GPT-OSS	1,169 ms	2,824 ms	1,111 ms	$0.261

Table 12: Latency and cost per model, seed-42 subset (30 questions × 9 conditions × 3 reps, n = 810 requests per model).

Condition	Mean input tokens	Cost / request
Missing evidence	382	$0.000250
Boundary	530	$0.000326
Conflict	779	$0.000451
Noise	996	$0.000548
Hard negative	1,202	$0.000646
Clean	1,542	$0.000817
Evidence position	1,546	$0.000817

Table 13: Mean input tokens and cost per request by condition, pooled across all five models (seed-42 subset). Each failure operator uses a fixed, operator-specific chunk count (missing evidence 1.3, boundary 3, conflict/hard negative 4, noise 5, clean/evidence position 8), so cost tracks operator design rather than a general "noise adds tokens" relationship; hard negative costs more per request than noise despite fewer chunks because its chunks average 1,208 characters versus 752 for noise.

E. Full Significance Test Tables

Condition	s42 (n=75)	s123+s2026 (n=131)	z	p-value
Missing evidence	0.787	0.809	−0.39	0.697
Conflict	0.280	0.977	−10.76	< 0.001
Hard negative	0.747	0.908	−3.12	0.002

Condition	s42 (n=75)	s123+s2026 (n=131)	z	p-value
Boundary	0.720	0.954	−4.79	< 0.001
Noise	0.653	0.969	−6.19	< 0.001
Evidence position	0.809	0.987	−7.97	< 0.001

Table 14: Two-proportion test comparing Qwen3-Next-80B-A3B's seed-42 accuracy against its pooled seed-123 and seed-2026 accuracy, per failure type.

Comparison	Condition	n (pairs)	A right / B wrong	A wrong / B right	χ2 (1 df)	p-value
Llama 3.3 vs. GPT-OSS	Hard negative	206 (3 seeds)	2	106	98.23	< 0.001
Llama 3.3 vs. DeepSeek V3.2	Hard negative	206 (3 seeds)	4	108	94.72	< 0.001
Nova Pro vs. DeepSeek V3.2	Conflict	206 (3 seeds)	0	40	38.03	< 0.001
Nova Pro vs. DeepSeek V3.2	Missing evidence	206 (3 seeds)	0	30	28.03	< 0.001

Table 15: McNemar's test on paired per-question correctness, pooled across all three benchmark seeds; continuity-corrected χ2, 1 degree of freedom.

Model	Mean failure accuracy (pooled)	95% CI (bootstrap, n = 206)
DeepSeek V3.2	0.930	[0.916, 0.945]
GPT-OSS	0.899	[0.880, 0.917]
Nova Pro	0.871	[0.852, 0.888]
Llama 3.3	0.829	[0.811, 0.846]

Table 16: Pooled bootstrap 95% confidence intervals for mean failure accuracy across all three seeds (5,000 resamples over 206 unique paired questions).

Judge agreement, parse rates, and the human-review queue: across all three seeds (9,270 judged instances), judge agreement ranges from 75.6% to 81.9%, with the remainder (≈18–24% per seed, ≈1,902 instances pooled) flagged for human review; 926 have been drawn into a review sample and adjudication is not yet complete for all of them. Structured-response parse rates average 93.3% (Llama 3.3) to 100% (DeepSeek V3.2), with corresponding low error rates (0–0.6%).

F. Detailed Evaluation Pipeline

Figure 1 (main text) summarizes the contribution at a glance; Figure 5 below shows the underlying eight-stage implementation. For each seed question, the pipeline draws a clean context and six paired failure contexts from RAGFailBench, constructs an identical prompt template around each context, issues the prompt to every model under fixed decoding settings, scores each response with rule-based checks (JSON validity, citation structure, abstention keyword), scores the same response with two independent LLM judges, escalating disagreements to human review, and aggregates the paired clean-versus-failure results into the metrics in Appendix A. Because only the retrieval context varies across conditions, any accuracy difference between a model's clean and failure-conditioned response on the same seed question is attributable to the injected failure rather than to question difficulty or prompt variation.

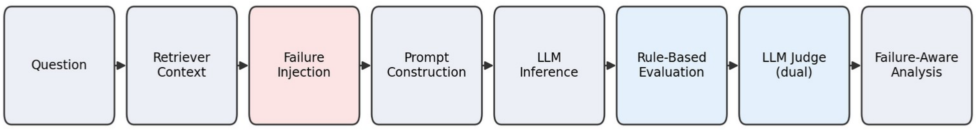

Figure 5: The eight-stage failure-aware evaluation pipeline underlying Figure 1. The failure-injection stage (highlighted) is the only point at which clean and failure-conditioned runs diverge for a given question.